

Topic: Data Definition Language (DDL) Commands

Q. Explain the various Data Definition Language (DDL) commands in SQL. Provide the syntax and a clear example for CREATE, ALTER, DROP, TRUNCATE, and RENAME.

Definition to Remember

Data Definition Language (DDL) is the subset of SQL used to define, modify, and delete the structural elements (schema) of a database — such as tables, views, and indexes. DDL commands are **auto-committed**, meaning changes are saved permanently and **cannot be rolled back**.

DDL Commands

1. CREATE

Used to create a new database or table.

```
CREATE TABLE Employee (  
    Emp_ID INT PRIMARY KEY,  
    Name   VARCHAR(50),  
    Salary DECIMAL(10,2)  
);
```

2. ALTER

Used to modify an existing table structure — add, delete, or modify columns.

```
-- Add a new column  
ALTER TABLE Employee ADD Department VARCHAR(30);  
  
-- Modify column size  
ALTER TABLE Employee MODIFY Name VARCHAR(100);
```

3. DROP

Used to **completely delete** a table — removes structure AND all data permanently.

```
DROP TABLE Employee;
```

 *Once dropped, the table cannot be recovered without a backup.*

4. TRUNCATE

Used to **delete all rows** from a table but **keeps the table structure** intact. Faster than DELETE because it does not log individual row deletions.

```
TRUNCATE TABLE Employee;
```

5. RENAME

Used to rename an existing table.

```
RENAME TABLE Employee TO Staff;    -- MySQL
-- Oracle: ALTER TABLE Employee RENAME TO Staff;
```

Summary Table

Command	Affects	Data Lost?	Auto-Committed?
CREATE	Creates table structure	N/A	Yes
ALTER	Modifies table structure	No	Yes
DROP	Deletes table + all data	Yes (completely)	Yes
TRUNCATE	Deletes all rows	Yes (data only)	Yes
RENAME	Renames table/column	No	Yes

★ Must-Write Points (for 10 marks)

1. DDL commands define the schema (structure) of a database.
2. All DDL commands are **auto-committed** — changes cannot be rolled back.
3. **CREATE**: creates a new table/database with specified columns and constraints.
4. **ALTER**: modifies existing table — add/drop/modify columns without losing data.
5. **DROP**: permanently deletes the entire table structure AND all its data.
6. **TRUNCATE**: deletes all rows but preserves the table structure; faster than DELETE.
7. **DROP** removes structure + data; **TRUNCATE** removes data only (structure remains).

⚡ Quick Recall

DDL → Auto-committed → CREATE (make table) → ALTER (modify structure) → DROP (delete all)
→ TRUNCATE (delete rows, keep structure) → RENAME (change name)